

Residential renewable energy adoption. Does financial literacy matter?

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ABSTRACT

Clean energy production systems are the most effective and sustainable ways of protecting the natural environment from pollution. Sadly, fossil fuels remain a significant energy source in most emerging economies primarily due to financial challenges. Using Ghana's renewable energy policy initiatives as a case analysis, the study adopted a multiple-stage random sampling approach to sample household heads (N = 413) across two regions in Ghana to assess the adoption of renewable energies' impact of financial literacy among urban households by applying the two-stage residual inclusion technique. The study's findings depict that household income and RE cost are more significant variables impacting RE adoption, whereas the focal study variable, i.e., financial literacy, has some positive effect but at a lower level than these two variables (household income and RE cost). The heterogeneous analysis regarding household income level shows that financial literacy increases the ability to adopt renewable energy in high-income households than low-income households, confirming the significant impact of household income on RE adoption. The study results imply that policymakers in low-income countries should provide tax incentives for importers of renewable energy products/appliances and subsidize the prices for consumers to improve renewable energy technology use. Also, the provision of job opportunities and income diversification strategies education to improve household incomes should be a priority. Moreover, financial literacy through financial education programs in the formal education sector and the mass media to improve citizens' financial skills and knowledge should be prioritized. These recommendations have the potential to advance the financial and economic wellbeing of citizens in facilitating renewable energy transition.

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Consent to participate

Participants consent were request and received.

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1. Introduction

As fossil fueled-powered energy generation systems have proven to be environmentally unsustainable and unreliable energy sources, the most effective and sustainable way of protecting the natural environment from dangerous pollutants such as CO₂ emissions is through clean energy production systems. Importantly, the rate at which climate change and energy volatility are increasing has drawn the attention of scholars and policymakers to low-carbon energy development as an ideal remedy for energy insecurity and global climatic warming mitigation (Asante et al., 2021; Bhattacharya et al., 2016). Sadly, in most emerging economies, fossil fuels and other traditional energies remain a significant source of energy needs. Renewable energy (RE) production presents a crucial opportunity for energy-poor countries to meet their energy demands while safeguarding the environment. However, financial challenges have rendered most RE policy initiatives in low-income economies stagnant.

Renewable energy advancement can reduce the overdependency of dirty fuel, which is causing environmental pollution, adverse health, and economic sluggishness, in developing countries like Ghana (Ankrah Twumasi, Jiang, Addai, et al., 2021; Liao et al., 2016). Also, clean energy production systems such as solar and biogas technologies bring the youth in developing countries up to speed with modern technological advancements and offer them avenues to build their skill sets to be more productive. As a result, several national RE policy interventions, such as Ghana's 10% clean energy production target in her total electricity mix by 2030, has been announced in many developing economies. Low-carbon energy transitions can help improve individuals' welfare, health outcomes, agriculture productivity, food security, and job creation (Ankrah Twumasi, et al., 2021; Hou et al., 2017; Williams et al., 2019). For example, Ankrah Twumasi et al. (2021) showed that dirty energy usage negatively affects household members' health status since emissions from fossil fuels promote diseases such as asthma, tuberculosis, chronic headache, and chronic pneumonia.

Additionally, harmful gases from fossil fuel use have contributed to extreme weather events such as droughts, heatwaves, and floods, causing colossal yield losses to the agriculture sector worldwide; hence, likely to cause food insecurity (Hou et al., 2017). Worryingly, the approximated figure (26.1 GtCO₂) of GHG emissions from fossil fuel combustion in 2004 is likely to hit 40GtCO₂ by 2030 if proper and rapid interventions to enhance RE evolutions are not strengthened (Gboney, 2009; IEA, 2019). Even though, Africa is not a significant source of GHG emissions in the world with a current share of 2%–3% from energy and industrial sources. However, African countries stand to bear the severe blunt of global warming and its consequent disasters such as food insecurity (famine), water deficits, and respiratory health challenges if innovative measures are not instituted to prevent a likely surge in her emission rates as projected in 2030. Based on these detrimental implications of dirty fuel production and the favorable implication of RE transitions discussed above, many governments and stakeholders are promoting renewable energy generation technologies through several policy interventions in recent times across national, regional, and continental levels. For example, in West Africa, the Centre for Renewable Energy and Energy Efficiency (ECREEE) has been established to superintend over the regional and national policy cohesion towards RE transition. However, the adoption of RE among households and firms in low-income countries, including Ghana, is very low mainly because of many barriers, including financial and economic barriers (Abolhosseini

and Heshmati, 2014; Asante et al., 2020).

Financial inclusion; thus, an individual's ability to participate in the financial market is regarded as one of the essential determinants of financial and economic standings among households (Ankrah Twumasi et al., 2020; Koomson and Danquah, 2021; Lahiani et al., 2021). It is, however, not surprising at the wider academic coverage of the significant relationship between financial services or financial market participation and households' livelihood improvement. Regarding this, access to financial services has been touted as a sure way to achieving most of the Sustainable Development Goals, including clean energy production for the sustainable energy for all (SE4ALL) commitment (Awaworyi Churchill et al., 2020). For example, Ankrah Twumasi et al. (2020) and Adusah-Poku and Takeuchi (2019) study showed that households who access financial services (e.g., received credit from financial intuitions) are more likely to increase their per capita household income; hence, affecting household clean energy consumption patterns. Household income enhancement promotes its members standard of living through good education, health services utilization, and clean energy and healthy meals consumption (Ankrah Twumasi et al., 2020; Lin et al., 2019; World Bank, 2017).

While financial inclusion has been one of the key contributors to household wellbeing, many adults (about 2 billion) in developing countries are still unbanked or not participating in the financial market (Demircuc-Kunt et al., 2017). Many challenging factors (including supply and demand-sides) attribute to this menace; however, low financial literacy is considered as one of the significant barriers at the demand side (Lusardi and Tufano, 2015; Van Rooij et al., 2011; Xu et al., 2019).

Financial literacy refers to individual's ability to conceptualize or develop economic and financial understanding and make knowledgeable decisions about financial planning, wealth accumulation, debt, and pensions management (Klapper and Lusardi, 2020). The definition suggests that with low financial literacy, individual's financial decisions and behaviors are inversely affected because of inadequate financial skills and knowledge. Those who are aware of financial services and products are individuals who seek out appropriate financial information to maximize their needs or incomes are more likely to be people who are financially literate (Askar and Quattara, 2020; Klapper and Lusardi, 2020; Lusardi and Mitchell, 2007). The reason is that questions like how, why, when and what to invest, borrow or save is highly connected to individuals' financial skills and knowledge, leading to sound financial decision-making. Therefore, financial market participation, an attitude to curb poverty and promote quality consumption patterns such as the installation of clean energy systems (e.g., off-grid solar energy systems) among households, is attainable through financial literacy. For example, Xu et al. (2019) and Ankrah Twumasi et al. (2022) showed that household consumption enhancement through access to financial services is highly influenced by financial literacy. These scenarios suggest that financial barriers to RE adoption can be curtailed by an income enhancer known as financial literacy. Thus, there is a potential association between financial literacy and RE adoption. However, a review of literature (e.g., Fong et al., 2021; Fornero et al., 2011; Lusardi, 2008; Murendo and Mutsonziwa, 2017; Zhang et al., 2021) related to financial literacy and household investment decision indicates an absence of data on whether households RE adoption can be promoted through financial literacy in developing countries like Ghana. We fill this gap using data from Ghana.

Ghana presents an interesting and relevant analytical lens to discuss. Ghana is a signatory to several international and regional treaties such as UN Sustainable Energy for all (SE4All), Paris agreement on carbon reduction, ECREEE and among others. Again, one of the country's energy policy priorities is to enhance the share of clean energy in the domestic energy mix according to the Energy Commission's Act 541 of 1997 (Energy Commission, 2018). For example, there is a 10% RE target from off-grid power systems agenda to be achieved by 2030. However, as of 2018, the nation could only boast of less than 3% RE sources from

off-grid power systems, with the remaining RE energy sources taking a record of hydro (40.79%) and thermal (57.21%) (Asumadu-Sarkodie and Owusu, 2016; Energy Commission, 2018). Although the country has a clean energy mix, it continues to experience perennial unreliable power supply because of growth in population, industrial expansion, technical anomalies and low rainfalls. Particularly, the low rainfalls make the two main hydro dams unable to produce enough power to meet the growing power demand (Asante et al., 2021). For example, the country experienced a hydropower generation shortage between 2007 and 2015 because of low rain showers (Asante et al., 2021; Eshun and Amoako-Tuffour, 2016). Meanwhile, the country has what it takes to generate RE sources from off-grid power systems. The bounteous RE sources (e.g., solar, biogas, tidal, geothermal, wind) gifted to the country has yet not been fully exploited. The average solar radiation enjoyed by the country stands at approximately 4–6 kWh/m²/day with insolation of 1800 h–3000 h p.a (Energy Commission, 2018); however, this solar potential has not been fully tapped. Therefore, policies to clear several impediments making the energy market in the country attractive to investors have become a policy priority.

To this end, the study's aims are twofold. First, we examined the RE adoption and financial literacy quantitative nexus and proposed a hypothesis that financial literacy can improve RE adoption. A financially literate persons, as mentioned previously, can make sound financial and investment decisions based on their acquired financial skills and knowledge to accumulate wealth and enhance household consumptions. Since inadequate funds have been one of the barriers to utilizing RE among households according to previous studies, and financial literacy can also improve household income, examining the RE adoption's effect of financial literacy (an income booster) becomes essential. Second, we investigate the heterogeneous impact of financial literacy in RE adoption by household income status. We augment existing literature in several ways. First, this is the first research work to explore the association between financial literacy and RE adoption in a developing country. Second, unlike previous RE adoption studies that prioritized loans, grants and, and government subsidies to address financial barriers to RE adoption (see e.g., Asante et al., 2020; Ghimire et al., 2018), we highlight the importance of addressing the possible role of financial literacy in enhancing clean energy consumption in developing countries. Third, the potential endogeneity problem linked to financial literacy is addressed using appropriate econometrics techniques. Most of the previous RE researchers failed to account for potential endogeneity problem in their study design, which is likely to cause inconsistencies and misleading research findings. However, this study's approach adequately dealt with potential endogeneity which could have affected the validity of the study conclusions.

The remaining parts of the study go as follows. Section 2 presents the study's method and material. While the results and discussion of the study are presented in Section 3, the conclusion and policy implication can be seen in Section 4.

2. Methods and material

2.1. Econometrics model

In this study, we assessed the relationship between financial literacy and renewable energy adoption. We argue that the focal variable, financial literacy, may be endogenous due to some reasons. First, financial literacy is not randomly distributed among the household heads. Thus, the household heads decide whether to seek financial knowledge and skills through consultation with expertise or participation in financial literacy programs to become financially literate. Also, householder/household factors likely to influence the household heads' financial literacy level may equally affect their renewable energy adoption rate. This implies that the financial literacy variable may be suffering from endogeneity issues that need to be addressed.

Looking at the nature of our outcome and focal variables (binary

variables) coupled with potential endogeneity problem that needed to be accounted for, we employed the two-stage residual inclusion (2SRI) technique. Terza et al. (2008) argued that the two-stage residual inclusion (2SRI) technique is useful in providing consistent results when dealing with binary variables with potential endogeneity problems. Thus, the 2SRI enables consistent estimates of the structural equation parameters because it considers the respondents' observed and unobserved attributes during the estimation process. With the 2SRI we can estimate the direct effect of financial literacy on RE adoption. Therefore, the two-stage residual inclusion (2SRI) technique is used for our analysis to abort inconsistency resulting from endogeneity issues in our results. Some scholars, including Ma and Zhu (2020), Kumar et al. (2020), and Nie et al. (2020), have applied the 2SRI approach in their studies. The maximum likelihood estimations of both first and second-stage equations are done separately using the 2SRI approach.

In stage one, the probability of being financially literate is estimated. Here, it is assumed that the i th respondent would develop his/her financial literacy capabilities through training programs and reading if the benefit or the expected utility derived from being financially literate outweigh the utility of being illiterate. The utility difference can be expressed by a latent variable as $FL_i^* = U_{i1}^* - U_{i0}^* > 0$; where U_{i1}^* and U_{i0}^* represent financial literate and illiterate, respectively. However, FL_i^* , which is a latent variable, is observable and takes the expression below:

$$FL_i^* = \alpha X_i + \beta I_i + \mu_i \dots FL_i = \begin{cases} 1 & \text{if } FL_i^* > 0 \\ 0 & \text{if, otherwise} \end{cases} \quad (1)$$

where FL_i^* refer to financial literacy's (FL) probability; i.e., $FL_i = 1$ if the household head is financially literate (whether the respondent score is above the median financial literacy score) and $FL_i = 0$ if otherwise; I_i is the instrumental variable (IV) in the first stage of the 2SRI model. Following Ankrah Twumasi et al. (2022), the instrument selected for the study is financial education (i.e., whether the household head has a relative/friend with an economics/financial education background). We followed X_i is a vector of control variables (see Table 1), α and β are the vector of parameters for the control variables which would be calculated and μ_i is a random disturbance term. Let us note that the financial education variable (IV) must be directly correlated with financial literacy but should not influence the outcome variable (renewable energy adoption) directly (Burgess et al., 2016). Therefore, it is a suitable instrument. In addition, we tested for its validity employing the Pearson correlation method (see Table A2 in appendix). In Table A1, financial literacy and adoption variables had a significant and insignificant correlation coefficient, respectively, suggesting that our IV is valid.

In stage two of 2SRI, the outcome variable is estimated by employing the Probit model because the outcome variable is a dummy variable (1 = adopters and 0 = non-adopters). The model is specified as follows;

$$REA_i^* = \gamma FL_i + \phi X_i + \varphi Residual_i + \varepsilon_i, \quad REA_i = \begin{cases} 1 & \text{if } REA_i^* > 0 \\ 0 & \text{if, otherwise} \end{cases} \quad (2)$$

where REA_i^* refers to the probability that a household head is a renewable energy adopter. It is represented by an observed binary indicator REA_i (1 = adopters and 0 = non-adopters); $Residual_i$ is a residual term predicted in the first stage, equation (1) estimation, and it is inserted in equation (2). This process helps address the issue of endogeneity that may cause inconsistency in the estimated outcome result (Ying et al., 2019). γ , ϕ , and φ are parameters to be estimated. ε_i is an error term.

2.2. Study data

The research data comes from a survey conducted in Ghana from June–August 2018. The study adopted a multiple-stage random sampling approach to sample household heads across two regions in Ghana.

Table 1
Socioeconomic, Demographic and other characteristics of the household heads.

Variables	Descriptions	Mean	SD
Dependent variables			
Adoption	Whether the household has adopted any renewable energy (1 = Yes, 0 = No)	0.27	0.42
Treatment variable			
Financial literacy	Whether the household head is financial literate (1 = Yes, 0 = No)	0.35	0.41
Independent variables			
Gender	Whether the household head is a male (1 = Yes, 0 = No)	0.62	0.53
Age	Age of the household head	44.14	11.27
Education	Household head formal education (in years)	12.21	7.51
Married	Household head is married (1 = Yes, 0 = No)	0.69	0.58
Household Dependency ratio	Number of older adults (60 years and above) and children below 12 years in a household	3.07	1.27
Internet use	Whether household head uses internet (1 = Yes, 0 = No)	0.79	0.58
Religious group	Whether the household head joins a religious group, i.e., Christian, Muslim, Traditionist (1 = Yes, 0 = No)	0.71	0.28
Cost perception	Whether a household head perceives that the cost of renewable energy use is expensive (1 = Yes, 0 = No)	0.62	0.53
Remittances	Whether the household head receives remittances (1 = Yes, 0 = No)	0.42	0.47
Climate change belief	Whether the household head believes in climate change (1 = Yes, 0 = No)	0.57	0.46
Ashanti region	Whether the household is in Ashanti (1 = Yes, 0 = No)	0.43	0.50
Greater Accra region	Whether the household is in Greater Accra (1 = Yes, 0 = No)	0.57	0.49
Financial education (IV)	Whether the household head has a relative/friend with economics/financial education background (1 = Yes, 0 = No)	0.43	0.47

Source: survey results 2018. During the survey in 2018, 1 USD = 5.2 Ghana cedis (GH¢)

Two regions, including Greater Accra and Ashanti, were selected in the first stage, randomly. These two regions were selected for some reasons. First, they are the most densely populated in the country, with a population of 4780380 (Greater Accra) and 4010054 (Ashanti) (GSS, 2015), representing 35.65%. While the Ashanti region serves as a destination for migrants from the northern and middle part of the nation, the Greater Accra is for those migrating in the southern sector. Again, according to statistics, all kinds of ethnic groups or tribes in Ghana can be located in these two regions (GSS, 2015). Based on this scenario, we are more convinced that the selection of these two regions is worthwhile. We further selected the capital cities in these regions in the second stage purposively; thus, Kumasi for Ashanti and Accra for Greater Accra. The focus was on the capital cities because it is easier to find renewable energy companies (e.g., installers, repairers, and appliance/product sellers). Also, these capital cities have variations in their socioeconomic and geographical attributes (Ghana Statistical Service, 2015b). A simple random approach was applied to pick roughly 30–40 households from the two cities in the last stage. After selecting 425 household heads for an interview, we used 413 of the administered questionnaires for our analysis because some of the questionnaires were not fully answered by the respondents.

Interview arrangements and pretested survey items were employed for the data collection. To make the respondents understand our questions, we did an in-depth face-to-face interview. Well-trained research assistants proficient in the local dialects assisted us in the interview section. The information gathered from the survey includes financial literacy assessment score, renewable energy adoption status, households' socioeconomic and demographical characteristics, perception of climate change, religious status, and other various determining factors

that will help achieve the aim of the study.

In this study, we asked the respondents to answer “yes” or “no” on the question; ‘Do you use any stand-alone renewable energy system to generate energy in your household?’ Two main RE sources emerged from our data (solar and biogas), and this is not surprising as these are the most popular off-grid RE sources in Ghana (Asante et al., 2021). We then group their answers into two renewable energy sources, namely, solar and biogas. A household head is considered an adopter if he/she uses any of the listed renewable energy sources to generate energy, suggesting that the adoption variable is binary; thus, 1 is equal to an adopter, and 0 equals otherwise. In terms of the financial literacy status, we developed a set of 7 questions following prior studies (for example, Andoh et al., 2015; Niu and Zhou, 2018; Xu et al., 2019) (see Table A1 in appendix). We obtained a possible total score ranging from 0 to 7 for every household head. These scores were transformed into dummies; that is, one (1) for a score higher than the median score of the total financial literacy score, and zero (0) for a score lower than the median score of the total financial literacy score. These researchers, including Wachira (2012) and Andoh et al. (2015) have applied this method in their financial literacy studies.

Also, based on prior studies (Ankrah Twumasi et al., 2021; Boogen et al., 2021; Murendo and Mutsonziwa, 2017; Putthinun et al., 2021; Van Rooij et al., 2011) and data available, other variables such as gender, age, years of schooling, marital, internet use, religious statuses and many others of the household/respondent were added as control variables. Table 1 exhibits all the study variables, their descriptions, means, and standard deviations. The data obtained from the survey were analyzed using STATA 15 and IBM SPSS version 26 statistical packages.

3. Results and discussions

3.1. Selected variables summary statistics

The explanations of the variables used in the analysis are displayed in Table 1. Most of the respondents (35%) are financially literate. Also, about 27% of the respondents had installed stand-alone RE systems. The percentages of male, married, internet users, and religious group affiliated respondents are 62%, 69%, 78%, and 71%, respectively. While about 62% of the respondents perceive that renewable energy adoption is expensive, 42% of the respondents are remittance receivers, and 57% believe in climate change. The household heads' average age and years of formal education are approximately 44 and 12 years, respectively. The average household dependency ratio of the sampled group is 3.07. About 43% and 57% of the households are located in the Ashanti and Greater Accra region, respectively. Finally, the sampled group depicts that 43% of the household heads have relatives or friends with economics/financial education backgrounds.

The average difference of the selected variables using illiterate and

Table 2
Mean differences in farmers attributes between financial literates and illiterates.

Variable	Literate	Illiterate	Differences
Adopters	0.33 (0.03)	0.20 (0.02)	0.12*
Gender	0.63 (0.05)	0.61 (0.05)	0.02
Age	49.78 (0.66)	38.50 (0.64)	11.28***
Education	17.03 (0.12)	7.39 (0.10)	9.64***
Married	0.67 (0.04)	0.71 (0.05)	−0.04
Household Dependency ratio	2.98 (0.08)	3.16 (0.11)	−0.18
Internet use	0.56 (0.04)	0.40 (0.02)	0.16**
Religious group	0.76 (0.06)	0.66 (0.05)	0.10*
Cost perception	0.63 (0.04)	0.61 (0.04)	0.02
Remittances	0.40 (0.02)	0.44 (0.03)	−0.04
Climate change belief	0.56 (0.04)	0.58 (0.04)	−0.02
Financial education (IV)	0.39 (0.02)	0.27 (0.01)	0.12***
Observation	146	267	Total = 413

Source: survey results, 2018. Note: ***p < 0.01, **p < 0.05, *p < 0.1. All numbers in parentheses are standard errors.

literate as a dimension can be seen in Table 2. The outcome variable, renewable energy adoption status, is very low for financially illiterate household heads, meaning the financially literates' renewable energy adoption is prevalent. This difference is in line with Fig. 1, which depicts that the adoption rate of RE for financially literate household heads from high-income and low-income households is higher than their financially illiterate counterparts. The results show that the young, educated, internet users, and religious group affiliates household heads are more likely to be financially literate than their illiterate counterparts. The analysis also shows that household heads having relatives/friends with economics/financial education backgrounds are financially literate. However, the results in Table 2 requires caution in its interpretation since it only shows a simple mean difference that considers not the householder/household observed and unobserved attributes. Therefore, we analyzed the financial literacy and renewable energy nexus quantitatively using an appropriate econometric technique (2SRI) that can bring consistency in our results.

3.2. Empirical results

We presented the impact of financial literacy on renewable adoption in Table 3. The Table shows that the residual value is significant, suggesting that there exist unobserved heterogeneities that can cause a bias in our analysis. Thus, using the 2SRI model for our analysis is appropriate and essential to account for the issue of endogeneity. We presented the financial literacy determinants in the second column of Table 3, while the third column presents renewable energy adoption determinates. Our study's discussion is based on the marginal effect results generated because of simplicity.

3.3. Determinants of financial literacy

Table 3, column 2 presents the results from the first stage regression, which focuses on the determinants of financial literacy. Specifically, we analyzed the effect of our instrument variable (i.e., financial education) on financial literacy. Our results indicate financial education (IV) had a positive and statistically significant effect on financial literacy. Empirically, we observe that household heads who relate with people with economic or financial education are 0.087 percent more likely to be financially literate compared to those with lower or no financial education. Household heads with such relatives or friends may seek financial advice from them to improve their financial skills and knowledge

Table 3

Renewable energy adoption effect of financial literacy (2SRI model).

Variables	First stage (Marginal effect)	Second stage (Marginal effect)
	Financial Literacy (0/1)	Renewable energy adoption
Financial Literacy		0.168 (0.061)***
Gender	0.018 (0.005)***	−0.108 (0.095)
Age	0.032 (0.011)***	0.016 (0.119)
Education	0.129 (0.062)***	0.049 (0.018)**
Married	0.009 (0.015)	0.303 (0.184)
Household Dependency ratio	−0.037 (0.055)	−0.028 (0.007)***
Internet use	0.029 (0.003)***	0.083 (0.051)
Religious group	0.057 (0.024)**	0.103 (0.086)
Cost perception	0.044 (0.091)	−0.022 (0.005)***
Remittances	−0.051 (0.037)	0.090 (0.057)
Climate change belief	0.016 (0.047)	0.041 (0.029)
Ashanti	0.022 (0.015)	−0.016 (0.007)*
Financial education (IV)	0.087 (0.030)***	
Residuals		0.112 (0.043)**
Log-pseudo likelihood	−1179.759	−1887.804
Regional dummies	Yes	Yes
Wald χ^2 (14)	59.79***	Prob > χ^2 = 0.0001
Observation	413	413

Source: survey results, 2018. Note: ***p < 0.01, **p < 0.05, *p < 0.1. All numbers in parentheses are robust standard errors. Greater Accra is the reference region.

necessary to allow them to make informed and effective financial choices compared to those with no financial education. This result is in line with the findings by Klapper and Lusardi (2020).

The marginal effect of gender indicates that male household heads are 0.018 percent more likely to be financially literate compared to their female counterparts. This result is expected because the male population is generally more represented in STEM programs such as mathematics and economics than females. As a results, males tend to do well in quantitative fields compared to females. This result is also supports Ankrah Twumasi et al. (2021) and Bucher-Koenen et al. (2016) findings, which showed that males are more financially literate than females. More so, the marginal effect of the age variable has a positive and 5 percent statistically significant effect on financial literacy. Our results suggest that one year increase in household head's age increases financial literacy by 0.032 percent. Expectedly, the marginal effect of the education variable has a positive and one percent statistically

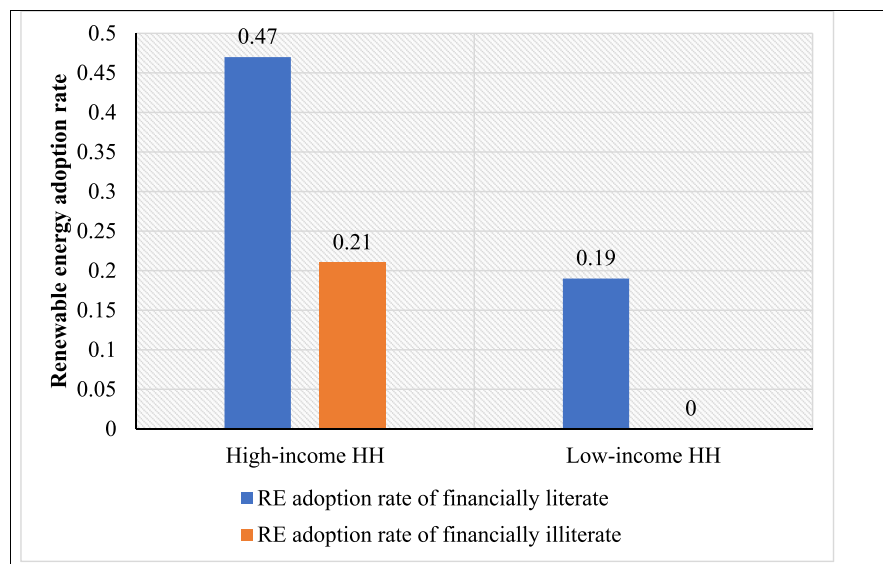


Fig. 1. Relationship between adopters and non-adopter RE adoption by household income levels. Note: HH= Household.

significant association with financial literacy. That is, household heads with higher levels of education are 0.129 percent more likely to be financially literate compared to household heads with few years of education. Thus, educational attainment not only develop the human capital of the household heads but also helps to improve their financial skills. This result is consistent with the study by Xu et al. (2019), who also found a positive and statistically significant effect of educational attainment and financial literacy.

The Interest use variable has a positive and statistically significant association with financial literacy. Our results indicate that household heads who use the Internet are about 0.029 percent more likely to be financially literate compared to their counterparts with no access to the Internet. Household heads that use the Internet tend to have access to variant of financial information, which improves their financial skills and knowledge and hence their financial literacy compared to household heads with no Internet use. Moreover, the marginal effect of the religious group variable has a positive and statistically significant association with financial literacy. Our result implies that household heads that join a religious group are 0.057 percent more likely to be financially literate than household heads who are not affiliated to any religious group. Thus, members in the religious group, for example, churches, learn various skills from each other. Also, sometimes churches invite resource persons from financial institutions to teach their members on savings, investments, planning, and how to manage personal finances, which affect their financial behavior.

3.4. The impact of financial literacy on renewable energy adoption

The second stage of our analysis examined the impact of financial literacy on renewable energy adoption. The results are presented in the third column of Table 3. Our results indicate a positive and one percent statistically significant marginal effect of financial literacy on renewable energy adoption. That is, we found that household heads who are financially literate are 0.168 percent more likely to adopt renewable energy compared to those who are financially illiterate. Household heads who are financially literate tend to patronize financial services such as borrowing and saving and have higher income, enabling them to consume energy efficiency equipment compared to those who are financially illiterate. Our result is consistent with Brent and Ward (2017), who analyzed the link between financial literacy and energy efficiency and found that a one standard deviation financial literacy increases the willingness-to-pay for energy efficiency by 10 percent.

Our results also revealed that household heads with higher education are 0.049 percent more likely to adopt renewable energy compared to household heads with no or low levels of education. Thus, those with education really understand the benefits associated with renewable energy use compared to household heads with no or low levels of education. The findings agree with Ma et al. (2019) and Rahut et al. (2016), who reported that educated people tend to use clean energy. More so, households with a higher dependency ratio are about 0.028 percent less likely to adopt renewable energy compared to households where the dependency ratio is less. This is because a high dependency ratio means low savings and income, making it impossible for the household to consume energy-efficient equipment (Awaworyi Churchill et al., 2020).

We also found a negative and statistically significant relationship between cost perception and renewable energy adoption. Empirically, household heads who perceived that the cost of renewable energy adoption is high are 0.022 percent less likely to adopt renewable energy compared to household heads with no cost perception of renewable energy. Individuals are sensitive to price and may be less willing to consume expensive products, all other things remaining constant (Goldsmith et al., 2010). Also, we found that household heads in the Ashanti region are less likely to adopt renewable energy compared to their counterparts in the Greater Accra region (the reference region). A plausible reason for this result can be attributed to spatial effects.

3.5. Further analysis

The association between financial literacy and renewable energy adoption is further explored using household income level as a baseline. The reason is that income plays a significant role in household goods and services demand, including renewable energy consumption. Households above(below) the median point of the household income are classified as high(low)-income households in this analysis. In the study, we posit that a high(low)-income household head who is financially literate is more (less) likely to increase the demand for household goods and services such as renewable energy use, especially in a developing country like Ghana. There is a high cost of renewable energy installation in Ghana due to the limited supply of renewable energy products and services (for example, inadequate installation expertise) (Asante et al., 2021). The disaggregated results are displayed in Table 4. It can be observed that financial literacy is positively associated with renewable energy adoption for both high-income and low-income households. However, our findings show no significant effect for low-income households. Generally, our results show that high-income households are the most beneficiaries in the financial literacy-renewable energy adoption positive association. This is evidence for policymakers in their pro-poor policy agenda.

3.6. Sensitivity analysis

Although we have employed the 2SRI model to estimate the quantitative relationship between financial literacy and renewable energy adoption, it is important to justify that our results are robust. Therefore, we performed a robustness test. In the robustness test analysis, we converted the respondents' financial literacy score from binary to a continuous variable; thus, the actual scores, which range from 0 to 7 was used. After, we employed the IV-probit model to estimate the quantitative relationship between financial literacy and renewable energy adoption rather than the 2SRI. The results obtained from the IV-probit (Table 5) analysis compared to the 2SRI have similarities; that is, both models show that a positive significant relationship exist between financial literacy and renewable energy adoption. The IV-Probit values are smaller because it does not capture unobserved attributes of the respondents. Even after using a different model and resetting the treatment variable, we observed that Tables 3 and 5 results are consistent. This suggests that the study's result is robust.

Table 4
Disaggregated estimation of financial literacy impact on renewable energy adoption by household income status 2SRI model.

Variable	Second stage (Marginal effect results)	
	High-income household	Low-income household
Financial Literacy	0.269 (0.021)***	0.198 (0.154)
Gender	0.013 (0.031)	-0.034 (0.028)
Age	0.028 (0.014)	-0.056 (0.021)**
Education	0.086 (0.038)*	0.034 (0.022)
Married	0.117 (0.212)	0.041 (0.039)
Household Dependency ratio	-0.042 (0.051)	-0.005 (0.021)
Internet use	0.089 (0.034)**	0.043(0.026)
Religious group	0.049 (0.085)	0.071 (0.095)
Cost perception	0.055 (0.049)	-0.036 (0.015)**
Remittances	0.013 (0.003)***	0.009 (0.017)
Climate change belief	0.017 (0.014)	0.061 (0.098)
Financial education (IV)	0.016 (0.011)	0.054 (0.040)
Residual	0.442 (0.062)***	0.091 (0.012)***
Log-pseudo likelihood	-2109.184	-1154.761
Regional dummies	Yes	Yes
Observation	195	218

Source: survey results, 2018. Note: ***p < 0.01, **p < 0.05, *p < 0.1. All numbers in parentheses are robust standard errors. The first stages are not presented because of brevity. Greater Accra is the reference region.

Table 5
Robustness checks results.

Variables	Renewable energy adoption (0/1) (IV Probit)
Financial literacy	0.076 (0.024)***
Control variables	Yes
Regional dummies	Yes
Instrumental variables	Yes
Wald χ^2	49.62***
Observation	413

Source: Survey results, 2018. Note: ***p < 0.01. All numbers in parentheses are robust standard errors. Reference region is Greater Accra.

4. Conclusion and policy implications

Household ability to install off-grid renewable energy technologies plays an important role in achieving the UN's sustainable energy development, clean production systems, and environmental goals. This paper explored financial literacy and clean energy generation and/or utilization nexus using primary data collected from two regions (urban households) in Ghana. Thus, we analyze how financial literacy can enhance household renewable energy adoption. This analysis is critical for sustainable energy production and environmental conservation in light of recent unparalleled climate change and energy volatility rates. According to our knowledge, there has not been any study addressing this issue in any developing country, including Ghana. Econometrics methods such as 2SRI and the IV-Probit as well as descriptive statistics, were used for the study's analysis. The conclusion emerging from our findings are as follows.

The estimated results showed that financially literate heads are more likely to install renewable energy systems privately than their illiterate counterparts. Thus, there was a positive nexus between financial literacy and RE adoption. Moreover, other independent variables, including education, household dependency, and cost perception, had a significant impact on renewable energy adoption. Also, the result from the disaggregated analysis of the household income showed that the financial literacy-renewable energy adoption relationship is positive for both low-income and high-income households. However, the magnitude of the effect was paramount among high-income households. Overall, the study's findings depict that household income and RE cost are more significant variables impacting RE adoption, whereas the focal study variable, i.e., financial literacy, has some positive effect but at a lower level than these two variables (household income and RE cost).

The results revealed the essence of providing some valuable policy implications. First, the positive relationship between financial literacy and renewable energy adoption indicates that financial education should be a policy priority and precede other clean production policy mechanisms. For instance, NGOs, policymakers, and governments can organize financial education programs in all national radio and television stations at least once a month to inculcate financial skills and knowledge to the citizens. These financial education programs can be localized at the district levels or in similar local political administrative units. Local political executives/administrators can liaise with financial experts and other resource persons to deliver financial literacy lectures to their constituents once a month. Also, clean energy adoption and cost perception had a negative relationship, implying that governmental interventions to reduce the cost of renewable energy products/appliances should be promoted. Governments should provide tax incentives for importers of renewable energy products/appliances to increase their

importation into the country and subsidize the prices for consumers to improve renewable energy use. When renewable products/appliances are in abundance, it will increase its suppliers' expansion in developing countries, expanding the renewable energy products market size. Also, the insignificant financial literacy-renewable association for low-income households calls for a pro-poor policy agenda in the country and similar economies to alleviate poverty and improve renewable energy adoption. Thus, the provision of job opportunities and income diversification strategies education to improve household incomes should be a priority for the governments. The attempt to increase renewable energy transition for electrification in the country could be achieved quickly by implementing these recommended policies. In sum, all these recommended policies mentioned above can be generalized for broader application in most developing countries, especially African countries, since they demonstrate similar economic development and socio-political outlook.

Some limitations emerge from this study that needs to be addressed. First, we used Ghana, a West African Country, as our sample domain due to inadequate funds, making it difficult to generalize our findings to all developing economies. However, since most African countries demonstrate similar economic development and socio-political outlook, we believe the observations in this study may be applicable in several other African economies. Therefore, future researchers can replicate this study in other developing countries in other parts of the globe. Moreover, the study evaluated only households' renewable energy adoption status. It is suggested that future scholars should consider other interesting avenues, such as the intensity of households' renewable energy adoption. It is one thing adopting and another thing intensifying the adoption. Finally, other political and social factors rather than financial literacy can influence renewable energy adoption in developing countries. Future studies can explore the relationship between those factors and renewable energy adoption to provide more alternative policies to help renewable energy transition and other cleaner production policy designs in developing nations.

CRediT authorship contribution statement

Martinson Ankrah Twumasi: writing, Writing – original draft, preparation, Investigation, writing, Writing – review & editing, Conceptualization, Methodology, Software, and, Formal analysis. **Den-nis Asante:** Conceptualization, Methodology. **Prince Fosu:** Supervision, Writing – review & editing. **Gloria Essilfie:** Writing – review & editing. **Yuansheng Jiang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1

Financial literacy questions and responses

Question	Answers
1. Suppose you had \$100 in your savings account with a 2% annual interest. After 5 years, how much will you have in this account if you leave your money to gain interest? (Interest rate)	(a) more than \$102 (b) exactly \$102 (c) less than \$102 (d) I do not know
2. Your savings account has a 1% annual interest, and the annual inflation is 2%. After a year, will you be able to buy more than today using the money saved in this account? (Inflation)	(a) Yes (b) No (c) I do not know
3. Is the following statement true or false? "Buying a single company stock usually provides a safer return than a stock mutual fund." (Risk diversification)	(a) True (b) False (c) I do not know
4. You want to borrow GH'500 from a moneylender (M1). He says that you can get it, but you must pay him GH'600 in a month. Another money lender (M2) says you have to pay GH'600 plus 15% in a month. Which one is better? (Borrowing)	(a) M1 (b) M2 (c) I do not know
5. Assume a friend inherits GH'10,000 today, and his sibling inherits GH'10,000 3 years from now. Who is richer because of the inheritance? (Time value of money)	(a) My friend (b) His sibling (c) They are equally rich (d) I do not know
6. Suppose that in the year 2010, your income has doubled, and the prices of all goods have doubled too. In 2010, how much will you be able to buy with your income? (Money illusion)	(a) More than today (b) The same (c) Less than today (d) I do not know
7. Second-hand farm machinery is more expensive to insure than a brand new one? (insurance)	(a) True (b) False (d) I do not know

Source: Ankrah Twumasi et al. (2021) and Lusardi and Mitchell (2007).

Table A2

Pearson correlation analysis of the selected IV

Variables	Correlation coefficient	p-value
Financial literacy	0.049**	0.016
RE adoption	0.186	0.112

Source: Survey results, 2018. Note: **p < 0.5.

References

- Abolhosseini, S., Heshmati, A., 2014. The main support mechanisms to finance renewable energy development. In: *Renewable and Sustainable Energy Reviews*, vol. 40. Elsevier, pp. 876–885.
- Adusah-Poku, F., Takeuchi, K., 2019. Household energy expenditure in Ghana: a double-hurdle model approach. In: *World Development*, vol. 117. Elsevier Ltd, pp. 266–277.
- Andoh, F.K., Nunoo, J., Darfor, K.N., 2015. Sustaining small and medium enterprises through financial service utilization: does financial literacy matter? *J. Bus. Entrep. Dev.* 5 (1), 74–94.
- Ankrah Twumasi, M., Jiang, Y., Addai, B., Asante, D., Liu, D., Ding, Z., 2021a. Determinants of household choice of cooking energy and the effect of clean cooking energy consumption on household members' health status: the case of rural Ghana. *Sustain. Prod. Consum.* 28, 484–495.
- Ankrah Twumasi, M., Jiang, Y., Adhikari, S., Adu Gyamfi, C., Asare, I., 2021b. Financial literacy and its determinants: the case of rural farm households in Ghana. *Agric. Finance Rev.* <https://doi.org/10.1108/AFR-06-2021-0078> available at.
- Ankrah Twumasi, M., Jiang, Y., Ameyaw, B., Danquah, F.O., Acheampong, M.O., 2020. The impact of credit accessibility on rural households clean cooking energy consumption: the case of Ghana. *Energy Rep.* 6, 974–983.
- Ankrah Twumasi, M., Jiang, Y., Wang, P., Ding, Z., Frempong, L.N., Acheampong, M.O., 2022. Does financial literacy inevitably lead to access to finance services? Evidence from rural Ghana. *Ciência Rural*. 52 (3) <https://doi.org/10.1590/0103-8478cr20210112> available at.
- Asante, D., He, Z., Adjei, N.O., Asante, B., 2020. Exploring the barriers to renewable energy adoption utilising MULTIMOORA- EDAS method. In: *Energy Policy*, vol. 142. Elsevier Ltd, 111479.
- Asante, D., He, Z., Mintah Ampaw, E., Gyamerah, S., Ankrah Twumasi, M., Opoku-Mensah, E., Kyere, F., et al., 2021. Renewable energy technology transition among small-and-medium scale firms in Ghana. *Renew. Energy* 178. <https://doi.org/10.1016/j.renene.2021.06.111> available at.
- Askar, M.W., Quattara, B., 2020. Financial literacy and poverty reduction: the case of Indonesia. *Ikonomika* 2 (1).
- Asumadu-Sarkodie, S., Owusu, P.A., 2016. The potential and economic viability of solar photovoltaic power in Ghana. *Energy Sources, Part A Recovery, Util. Environ. Eff.* 38 (5), 709–716.
- Awaworyi Churchill, S., Smyth, R., Farrell, L., 2020. Fuel poverty and subjective wellbeing. *Energy Econ.* 86 <https://doi.org/10.1016/j.eneco.2019.104650> available at.
- Bhattacharya, M., Paramati, S.R., Ozturk, I., Bhattacharya, S., 2016. The effect of renewable energy consumption on economic growth: evidence from top 38 countries. In: *Applied Energy*, vol. 162. Elsevier Ltd, pp. 733–741.
- Boogen, N., Cattaneo, C., Filippini, M., Obrist, A., 2021. Energy efficiency and the role of energy-related financial literacy: evidence from the European residential sector. *Energy Effic.* 14 (4), 40.
- Brent, D.A., Ward, M., 2017. Energy efficiency and financial literacy energy efficiency and financial literacy working paper. *J. Environ. Econ. Manag.* 90.
- Bucher-Koenen, T., Lusardi, A., Alessie, R.J.M., Van Rooij, M.C.J., Washington, G., 2016. How financially literate are women? an overview and new insights. *J. Consum. Aff.* 51, 255–283. <https://doi.org/10.1111/joca.12121> available at. <http://www.nber.org/papers/w20793>.
- Burgess, S., Dudbridge, F., Thompson, S.G., 2016. Combining information on multiple instrumental variables in Mendelian randomization: comparison of allele score and summarized data methods. *Stat. Med.* 35 (11), 1880–1906.
- Demirgüç-Kunt, A., Singer, D., April 25, 2017. Financial Inclusion and Inclusive Growth: A Review of Recent Empirical Evidence. World Bank Policy Research Working Paper No. 8040. Available at SSRN: <https://ssrn.com/abstract=2958542>.
- Energy Commission, 2018. *Energy (Supply and Demand) Outlook for Ghana*, vol. 5. Ghana Energy Commission, Accra, Ghana.
- Eshun, M.E., Amoako-Tuffour, J., 2016. A review of the trends in Ghana's power sector. *Energy, Sustain. Soc.* 6 (1), 9.
- Fong, J.H., Koh, B.S.K., Mitchell, O.S., Rohwedder, S., 2021. Financial literacy and financial decision-making at older ages. *Pac. Basin Finance J.* 65 <https://doi.org/10.1016/j.pacfin.2020.101481> available at.
- Fornero, E., Monticone, C., Trucchi, S., 2011. The effect of financial literacy on mortgage choices. SSRN. available at. <https://doi.org/10.2139/ssrn.1950040>.
- Gbone, W., 2009. Policy and regulatory framework for renewable energy and energy efficiency development in Ghana. *Clim. Pol.* 9, 508–516. <https://doi.org/10.3763/cpol.2009.0636> available at.
- Ghana Statistical Service, 2015a. "Annual Gross Domestic Product", Ghana Statistical Service: Statistics for Development and Progress.
- Ghana Statistical Service, 2015b. *Ghana Poverty Mapping Report*.

- Ghimire, L.P., et al., 2018. An analysis on barriers to renewable energy development in the context of Nepal using AHP. *Renew. Energy* 129, 446–456. <https://doi.org/10.1016/j.renene.2018.06.011>.
- Goldsmith, R.E., Flynn, L.R., Kim, D., 2010. Status consumption and price sensitivity. *J. Market. Theor. Pract.* 18 (4), 323–338.
- Hou, B.D., Tang, X., Ma, C., Liu, L., Wei, Y.M., Liao, H., 2017. Cooking fuel choice in rural China: results from microdata. In: *Journal of Cleaner Production*, vol. 142. Elsevier Ltd, pp. 538–547.
- IEA, 2019. *Renewables 2019 – Analysis* - IEA. International Energy Agency.
- Klapper, L., Lusardi, A., 2020. Financial literacy and financial resilience: evidence from around the world. *Financ. Manag.* 49 (3), 589–614.
- Koomson, I., Danquah, M., 2021. Financial inclusion and energy poverty: empirical evidence from Ghana. *Energy Econ.* 94 <https://doi.org/10.1016/j.eneco.2020.105085>. No. January, available at.
- Kumar, A., Mishra, A.K., Saroj, S., Thapa, G., Joshi, P.K., 2020. Food safety measures and food security of smallholder dairy farmers : empirical evidence from Bihar , India. *Agribusiness* 36 (3), 363–384.
- Lahiani, A., Mefteh-Wali, S., Shahbaz, M., Vo, X.V., 2021. Does financial development influence renewable energy consumption to achieve carbon neutrality in the USA? *Energy Pol.* 158, 112524.
- Liao, H., Tang, X., Wei, Y.M., 2016. Solid fuel use in rural China and its health effects. *Renew. Sustain. Energy Rev.* 60, 900–908. <https://doi.org/10.1016/j.rser.2016.01.121> available at.
- Lin, L., Wang, W., Gan, C., Nguyen, Q.T.T., 2019. Credit constraints on farm household welfare in rural China: evidence from fujian province. *Sustainability* 11 (11), 3221.
- Lusardi, A., 2008. Household Saving Behavior : the Role of Financial Literacy , Information , and Financial Education Programs (Dartmouth College and NBER) February 2008", *Education*, vol. 1, pp. 1–43. February.
- Lusardi, A., Mitchell, O.S., 2007. Financial literacy and retirement preparedness: evidence and implications for financial education programs. *Bus. Econ.* 42, 35–44. <https://doi.org/10.2145/20070104> available at.
- Lusardi, A., Tufano, P., 2015. Debt literacy, financial experiences, and overindebtedness. *J. Pension Econ. Finance* 14, 332–368. <https://doi.org/10.1017/S1474747215000232> available at.
- Ma, W., Zhou, X., Renwick, A., 2019. Impact of off-farm income on household energy expenditures in China: implications for rural energy transition. In: *Energy Policy*, vol. 127. Elsevier Ltd, pp. 248–258. December 2018.
- Ma, W., Zhu, Z., 2020. A note: reducing cropland abandonment in China – do agricultural cooperatives play a role? *J. Agric. Econ.* 71 (3), 929–935.
- Murendo, C., Mutsonziwa, K., 2017. Financial literacy and savings decisions by adult financial consumers in Zimbabwe. *Int. J. Consum. Stud.* 41 (1), 95–103.
- Nie, P., Ma, W., Sousa-Poza, A., 2020. The relationship between smartphone use and subjective well-being in rural China. In: *Electronic Commerce Research*. Springer US. <https://doi.org/10.1007/s10660-020-09397-1> available at.
- Niu, G., Zhou, Y., 2018. Financial literacy and retirement planning: evidence from China. *Appl. Econ. Lett.* 25 (9), 619–623.
- Putthinun, P., Watanapongvanich, S., Khan, M.S.R., Kadoya, Y., 2021. Financial literacy and alcohol drinking behavior: evidence from Japan. *Sustainability* 13 (16). <https://doi.org/10.3390/su13168858> available at.
- Rahut, D.B., Behera, B., Ali, A., 2016. Household energy choice and consumption intensity: empirical evidence from Bhutan. In: *Renewable and Sustainable Energy Reviews*, vol. 53. Elsevier, pp. 993–1009.
- Van Rooij, M., Lusardi, A., Alessie, R., 2011. Financial literacy and stock market participation. In: *Journal of Financial Economics*, vol. 101. Elsevier, pp. 449–472, 2.
- Terza, J.V., Basu, A., Rathouz, P.J., 2008. Two-stage residual inclusion estimation: addressing endogeneity in health econometric modeling. *J. Health Econ.* 27 (3), 531–543.
- Wachira, M.I., 2012. Impact of Financial Literacy on Access to Financial Services in Kenya, vol. 3, pp. 42–50, 19.
- Williams, P.A., Crespo, O., Abu, M., 2019. Adapting to changing climate through improving adaptive capacity at the local level – the case of smallholder horticultural producers in Ghana. *Clim. Risk Manag.* 23, 124–135.
- World Bank, 2017. Monitoring global poverty report of the commission on global poverty monitoring global poverty: report of the commission on global poverty advises the World Bank on the measurement and monitoring of global poverty in two areas. available at. <https://openknowledge.worldbank.org/bitstream/handle/10986/25141/9781464809613.pdf>.
- Xu, N., Shi, J., Rong, Z., Yuan, Y., 2019. Financial literacy and formal credit accessibility: evidence from informal businesses in China. In: *Finance Research Letters*. Elsevier Inc., 101327.
- Ying, A., Xu, R., Murphy, J., 2019. Two-stage residual inclusion for survival data and competing risks—an instrumental variable approach with application to SEER-Medicare linked data. *Stat. Med.* 38 (3), 1775–1801.
- Zhang, Y., Jia, Q., Chen, C., 2021. Risk attitude, financial literacy and household consumption: evidence from stock market crash in China. *Econ. Modell.* 94 <https://doi.org/10.1016/j.econmod.2020.02.040> available at.